

LAB ASSIGNMENT-3

Title: Simple and Multi-Linear Regression

- Implement simple and multi-linear regression to predict profits for a food truck. Compare the performance of the model on linear and multi-linear regression.

Objective: The objective of this lab assignment is to implement simple and multi-linear regression models to predict profits for a food truck business. By comparing the performance of these two regression models, you will gain insights into when and how to use simple and multi-linear regression techniques.

Dataset Format:

Population	Years in business	Profit
10,000	5	10000
15000	6	12000
20000	6	13000
9000	5	12000
12000	4	?

Tasks:

- 1) Apply Simple Linear Regression.
- 2) Implement **Normal Equations** and **Gradient Descent** manually.
- 3) Performance Evaluation (Simple Linear Regression).
- 4) Multi-Linear Regression.
- 5) Performance Evaluation (Multi-Linear Regression).
- 6) Model Comparison and Interpretation using **R^2** , **MSE**, **MAE**, **RMSE**.

Submission:

Prepare a PDF file that covers all the tasks mentioned above. Include code snippets, visualizations, and tables to support your analysis. Clearly explain the steps you took, the results you obtained, and your interpretation of the findings.